followed in this project to facilitate easy debugging of various modules. ModelSim is an easy-to-use yet versatile simulator from Mentor Graphics. It supports behavioural, register-transfer- and gate-level modelling.

To start memory (64-bit) design simulation, install ModelSim V 10.4a on a Windows PC and follow the steps mentioned below:

1. Start ModelSim from the desktop; you will see ModelSim10.4a dialogue window

2. Create a project by clicking Jump Start on Welcome screen

3. Create Project window pops up (Fig. 2). Select a suitable name for your project. Set project location and leave the rest as default, and click OK

4. Add items to Project window pops up (Fig. 3). Select Create New File option

5. Create Project File window pops up. Give an appropriate file name (say, Memory_SP.v) for the file you want to add, and choose Verilog as Add file as type and Top Level as Folder (Fig. 4)

6. On the workspace section of the main window (Fig. 5), double-click on the file you have just created (Memory_SP.v in this case)

7. Type your Verilog code (Memory_SP.v) in the new window. Since 64-bit x 8-bit memory is considered here, 64-bit address is considered, and the main goal is to input 8-bit data that should be written in this address location. Note that these inputs are arbitrary and you can enter any 8-bit data

8. Save your code from File menu

Note. Generally, memory internal architecture design may contain D-flip-flop, mux, basic cell, 8-bit word, core and so-on.

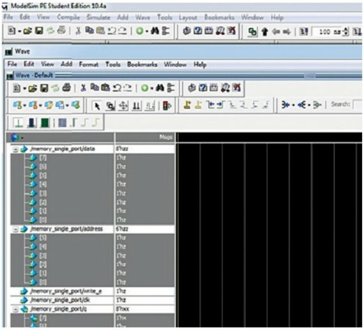

Fig. 10: Selecting and monitoring signals

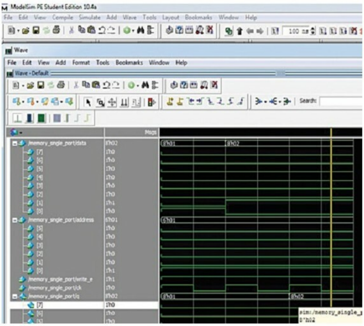

Fig. 11: Wave window

Compiling/debugging project files

1. Select Compile→Compile All options

2. Compilation result is shown on the main window. If you get 'Compile of memory-SP.V was successful' message, it means there are no errors in the project (Fig. 6)

Simulating memory 64-bit×8-bit design

1. Click on Library menu from the main window and then click on plus (+) sign next to the work library. You would see Memory_SP that you have just compiled (Fig. 7)

2. In the work library, select as shown in Fig. 7 and click OK. This will open sim-Default window, as shown in Fig. 8

3. Go to Add→To Wave→All items in design options (Fig. 9)

4. Select the signals you want to monitor for simulation purposes. Select these, as shown in Fig. 10

5. Selected signals are used to verify data-write operations in the memory. Simulate your design by clicking Run from Simulate menu bar, as shown in Fig. 11

Here, to validate the memory implementation, fill the memory at selected address location between 0 and 63. Perform a random data write operation. Enable and read the data from that particular address location. EFY

EFY Note
The source code of this project is included in this month's EFY DVD and is also available for free download at *source.efymag.com*

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